TELE-ICU SMART ASSET & REMOTE CALIBRATION NETWORK

A PROJECT REPORT

Submitted by

SHREMATHI K

(2116240701504)

in partial fulfilment of the End Semester Examination for the course

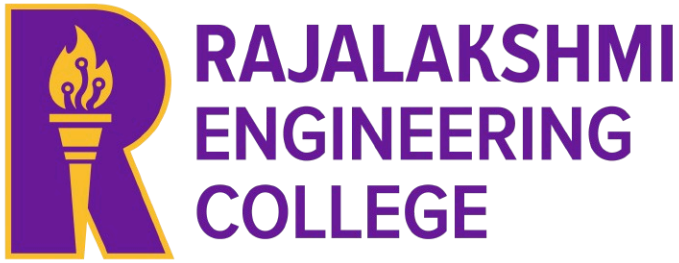
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

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RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled “Tele-ICU Smart Asset & Remote Calibration Network” is the bonafide work of Shremathi K (2116240701504), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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Submitted for the End Semester Examination (Individual Project) held on 11.11.2026.

INTERNAL EXAMINER	**EXTERNAL EXAMINER**

DECLARATION

I, Shremathi K (2116240701504), hereby declare that the project report titled “Tele-ICU Smart Asset & Remote Calibration Network”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai Date: 11.11.2006	**SIGNATURE OF THE STUDENT** SHREMATHI K 2116240701504

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SHREMATHI K

ABSTRACT

Hospitals depend on a wide range of medical devices — ventilators, monitors and other ICU equipment — that are distributed across several departments and must be calibrated, maintained and serviced at regular intervals. When asset records, calibration schedules, maintenance requests and spare-parts information are held in separate registers or disconnected files, it becomes difficult to maintain a single view of equipment, to notice calibration deadlines that are approaching or overdue, and to trace the service history of a device.

This project presents “Tele-ICU Asset Management”, a low-code application titled the Tele-ICU Smart Asset & Remote Calibration Network, developed on the Zoho Creator platform for the centralised management of medical equipment across hospitals and ICU departments. The application comprises forms for hospitals, ICU departments, technicians, a medical asset registry, calibration management, maintenance requests, maintenance work orders, spare-parts inventory, asset transfers and service histories, together with an approval request form. Subforms record calibration checklists and maintenance task checklists. Deluge scripting and workflows monitor calibration due dates, send a daily calibration reminder scheduled for 9:00 AM, and send an email notification when a maintenance request is submitted. Reports, an overview page and dashboards present asset status and calibration information, and a role hierarchy with a data-sharing rule allows calibration records created by a Manager to be viewed in read-only mode by the CEO.

The core application has been developed and verified through functional testing using sample records: asset records were created and viewed in reports, lookup fields were exercised, the calibration deadline script was executed manually and its reminder email was received, and a test maintenance request produced the expected email notification. Details of deployment, performance and security evaluation, and user acceptance testing are working . The work demonstrates that a coherent equipment-management solution can be assembled rapidly on a low-code platform with limited custom code.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Role-Based Access Control; Medical Asset Management.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Tele-ICU Smart Asset & Remote Calibration Network
Application Domain	Healthcare, hospital equipment management, and low-code application development
Student Name & Register No.	Shremathi K – 2116240701504
Zoho Creator Edition	Zoho Creator Free/Trial Edition
Application Name	Tele-ICU Asset Management
Application Link Name	Tele-ICU Asset Management
Published / Live URL	Not yet published
Demo Credentials (per role)	CEO: Manager:
Number of Forms / Subforms	**11 forms/modules documented** *(including Approval Request; confirm final count)* / **2 subforms:** Calibration Checklist and Maintenance Task Checklist
Number of Reports (by type)	Exact count and types 12
Number of Pages / Dashboards	**3:** Tele-ICU Overview, Tele-ICU Dashboard, Approval Dashboard
Number of Workflows	3 documented: Calibration_Deadline_Monitor, Daily_Calibration_Reminder (Schedule), Maintenance_Request_Notification.
Deluge Functions / Custom Actions	Deluge scripts used in Calibration Deadline Monitor and Daily Calibration Reminder for checking calibration due dates and sending email alerts.

Integrations	Email notifications only; no other Zoho service or external API integration documented
Roles & Profiles Defined	CEO, Manager (Manager reports to CEO)
Environments Used	Zoho Creator (Low-Code Application Development Platform), Deluge Scripting, and Zoho Creator's built-in form, report, workflow, and dashboard tools.
Date of Production Deployment	Not yet deployed to production

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CMMS	Computerised Maintenance Management System
CRUD	Create, Read, Update, Delete
Deluge	Data Enriched Language for the Universal Grid Environment
ER	Entity–Relationship
FR	Functional Requirement
ICU	Intensive Care Unit
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
SDLC	Software Development Life Cycle
UAT	User Acceptance Testing
UI	User Interface
WHO	World Health Organization

CHAPTER 1

INTRODUCTION

1.1 Background

Hospitals and intensive care units rely on a large and varied inventory of medical equipment, including ventilators, patient monitors and other devices that support patient care. Such equipment represents a significant investment and must remain safe, reliable and available whenever it is needed. The World Health Organization describes a maintenance strategy for medical equipment as one that combines procedures for inspection with preventive and corrective maintenance [1]. Supporting this strategy requires an accurate inventory of what equipment exists and where it is located [2], and the use of a computerised system to record maintenance activities is a recognised means of managing the information involved [3].

In practice, equipment is distributed across several departments and, in larger organisations, across more than one hospital. Each device has its own calibration interval, service requirements and maintenance history. Technicians must be assigned to repair and service tasks, spare parts must be available when a device is repaired, and equipment is sometimes moved from one department to another. All of these activities generate records that must be kept consistent and must be visible to the people who are responsible for them.

This project, titled the Tele-ICU Smart Asset & Remote Calibration Network, addresses this process through a low-code application named “Tele-ICU Asset Management”, developed on the Zoho Creator platform. The application provides a centralised location in which hospitals, ICU departments, technicians and medical assets are recorded, and in which calibration schedules, maintenance requests, work orders, spare parts, asset transfers and service histories are managed. The organisation for which the application is intended, its scale, and the process currently followed. The records used while developing and testing the application are sample test records only, such as a sample hospital named “Chennai Central Hospital” with the code HSP001, and do not represent real hospital data.

It should be noted that the term “Remote Calibration” in the project title refers to the centralised and remote visibility of calibration schedules, due dates and reminders across hospitals and departments. The application records and monitors calibration information; it does not connect to medical devices to perform calibration automatically.

1.2 Problem Statement

Hospitals use different medical devices across multiple departments. These devices require regular calibration, maintenance, servicing and accurate record keeping. Manual or disconnected processes make it difficult to maintain centralised asset records, to monitor calibration deadlines, to identify overdue equipment, to track maintenance requests and work orders, and to notify the relevant users. This project addresses these challenges through a centralised low-code application for medical asset, calibration and maintenance management. The specific pain points are as follows.

- Asset information is not held in a single centralised registry, making it difficult to know which equipment exists in which hospital or ICU department.
- Calibration deadlines are difficult to monitor, and equipment whose calibration is overdue or approaching its due date is not easily identified.
- Maintenance requests are not tracked consistently, and the assignment of work orders and the completion of maintenance tasks are difficult to follow.

- Service history, spare-parts availability and asset transfers are not recorded in a connected manner.
- Relevant users are not notified promptly of calibration deadlines or newly submitted maintenance requests.

1.3 Objectives

The objectives of the project are as follows. Each objective is mapped to the implementation in Chapter 5 and to the results in Chapter 8 (Table 8.1).

- To create a centralised medical asset registry, together with hospital, ICU department and technician records, using a structured data model of Zoho Creator forms and lookup fields.
- To track calibration schedules and due dates, and to maintain calibration checklist records, using forms and a subform.
- To submit and track maintenance requests, and to manage maintenance work orders and task checklists.
- To maintain spare-parts inventory, record medical asset transfers, and maintain service and maintenance history.
- To automate calibration deadline monitoring and email notifications using Deluge scripting and workflows.
- To provide reports and dashboards for asset and calibration information, and to configure user roles and permissions for controlled access to the data.

1.4 Scope of the Project

In scope: The project covers the design and development of the Tele-ICU Asset Management application on Zoho Creator. The modules developed or documented are Hospital Management, ICU Department Management, Technician Management, Medical Asset Registry, Calibration Management, Maintenance Requests, Maintenance Work Orders, Parts Inventories (Spare Parts Inventory), Asset Transfer Management, Service Histories and an Approval Request form. The application includes two subforms (Calibration Checklist and Maintenance Task Checklist), reports, an overview page, dashboards, three workflows, Deluge scripting, email notifications, and two user roles (CEO and Manager) with read and write permissions and a data-sharing rule. The application is accessed through a web browser.

Out of scope: The application does not implement SMS, WhatsApp or push notifications; email is the only notification channel. It does not connect directly to medical devices, and no integration with other Zoho services or external APIs is documented in this report. A customer or patient portal is not part of the project.

1.5 Justification for a Low-Code Approach

Low-code development platforms provide visual environments in which applications are assembled largely through configuration, so that functioning software can be produced without writing extensive programs [6]. A study of practitioner discussions on Stack Overflow and Reddit reports that such platforms offer drag-and-drop interfaces, supply ready-made units such as components and APIs that make them easy to learn and speed up development, and are particularly favoured where automated processes and workflows are required [4]. A comparative study of low-code platforms similarly describes them as cloud-delivered environments that allow citizen developers to build applications without the help of a large team of programmers [5].

These characteristics suit the present problem. Medical asset management is a data- and process-oriented problem: it consists of structured records, relationships between them (assets belong to hospitals and departments), scheduled checks and notifications, and role-based access to data. Zoho Creator provides forms, subforms, lookup fields, reports, pages, workflows, user permissions and the Deluge scripting language as built-

in capabilities [7]. The platform also hosts the application, so no separate server, database or deployment pipeline had to be built. Custom code was therefore limited to the areas where date evaluation and dynamic email content were required, namely the calibration monitoring and reminder logic. The same application, if developed with traditional programming, would have required the design and hosting of a database, an application server, a user interface and a notification mechanism before any business logic could be implemented.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The process that the project replaces is described in terms of the problem statement of Section 1.2: equipment records are maintained through manual or disconnected means. In such a process, the typical sequence of activities is as follows. Equipment details are recorded when a device is acquired and installed. The calibration date and next due date are noted separately, and someone must remember or periodically check which devices are due. When a device develops a fault, a maintenance request is communicated to the maintenance personnel, a technician is assigned, spare parts are obtained if required, and the work performed is noted. If a device is moved to another department, the change is recorded separately, and the service history of the device has to be reconstructed from several sources whenever it is required.

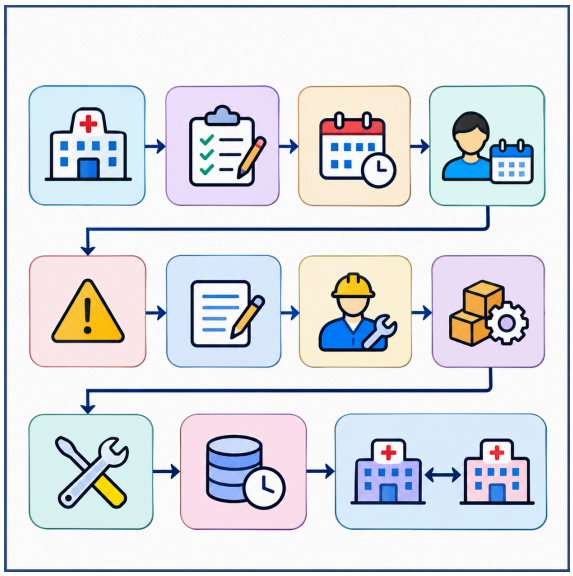


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Asset records are not centralised, so the equipment of different hospitals and departments cannot be viewed together.
- Calibration due dates must be tracked manually, which makes overdue and upcoming calibrations easy to miss.
- Maintenance requests and work orders are not tracked through a single, consistent record.
- Service history, spare-parts inventory and asset transfers are kept separately and are difficult to relate to one another.
- Users are not notified automatically about deadlines or new maintenance requests.
- There is no controlled, role-based access to equipment information.

2.3 Low-Code versus Traditional Development

Table 2.1 compares the two approaches for the proposed application. Entries that depend on measurements made by the student are left for the student to complete; the remaining entries describe the general nature of each approach.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Estimated effort: **6–8 weeks**	Actual effort: **3–4 weeks**
Skills required	Programming, database design, web front-end and back-end development	Data modelling, form and report configuration, Deluge scripting
Hosting & infrastructure	Servers, database and deployment to be provisioned and maintained	Provided by the Zoho Creator platform
Mobile application	Separate mobile development or a responsive web design	Zoho Creator mobile app availability: **Available for Android and iOS**
Security & access control	Authentication and authorisation to be designed and coded	Roles, permissions and sharing rules configured on the platform
Maintenance & change	Code changes, testing and redeployment	Changes made in the form builder, workflow and script editors
Customisation limits	Few limits other than time and cost	Bound by platform features and Deluge limits (see Section 8.4)
Cost	Development, infrastructure and maintenance costs: **₹40,000–₹60,000 (estimated)**	Platform licence/edition cost: **₹0 during the complimentary trial; paid plan required for continued use**

2.4 Literature Survey

Table 2.2 summarises the sources reviewed. They fall into two groups: studies of low-code development platforms [4], [5], [8], and guidance on the management of medical equipment [1]–[3]. The Zoho Creator documentation [7] and Deluge reference [9] were used for platform capabilities.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Y. Luo et al., 2021 [4]	Characteristics and Challenges of Low-Code Development: The Practitioners' Perspective	Analysis of Stack Overflow and Reddit posts found that low-code offers drag-and-drop development, ready-made units, faster development, suitability for process and workflow automation, and conflicting practitioner views on advantages and disadvantages.	Supports the choice of low-code for a workflow-oriented application and the discussion of limitations.
2	A. Sahay et al., 2020 [5]	Supporting the understanding and comparison of low-code development platforms	Presents a layered view of low-code platforms and a comparison framework to help users choose a platform that suits their requirements.	Provides the conceptual basis for the platform and architecture discussion.
3	World Health Organization, 2011 [1]	Medical equipment maintenance programme overview	A maintenance strategy includes procedures for inspection, preventive maintenance and corrective maintenance.	Defines the maintenance activities that the application records and tracks.
4	World Health Organization, 2011 [2]	Introduction to medical equipment inventory management	Addresses inventory management of medical equipment as a basis for maintenance.	Supports the need for a centralised medical asset registry.
5	World Health Organization, 2011 [3]	Computerized maintenance management system	Part of the WHO technical series on computerised systems for managing equipment maintenance.	Related category of product to which this application belongs.
6	U. Frank et al., 2021 [8]	Low Code Platforms: Promises, Concepts and Prospects – A Comparative Study of Ten Systems	Studies ten platforms and notes that there is no unified definition of low-code.	Supports the positioning of the application within the low-code spectrum.

7	Zoho Corporation [7]	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows, roles and Deluge scripting.	Primary reference for the implementation.

2.5 Emerging Trends in Low-Code Development

Several trends in low-code development are relevant to this project. First, practitioners report that low-code platforms are especially favoured in domains that need automated processes and workflows [4]; the calibration monitoring and notification workflows of this application are an example. Second, the participation of non-specialist “citizen developers” in building applications is a central promise of low-code platforms [5], which is consistent with the way this application was assembled from forms, reports and pages. Third, comparative studies emphasise that platforms differ in their features and that the choice of platform should follow the requirements of the problem [5], [8]. Finally, AI-assisted application building is an emerging feature of several platforms.

2.6 Proposed System

The proposed system is the Tele-ICU Asset Management application on Zoho Creator. It overcomes the limitations of Section 2.2 as follows. The Medical Asset Registry, linked by lookup fields to the Hospital Management and ICU Department Management forms, centralises asset records. Calibration dates are stored in the registry and in Calibration Management, and the Calibration_Deadline_Monitor and Daily_Calibration_Reminder workflows identify overdue and upcoming calibrations and send reminder emails. Maintenance Requests, Maintenance Work Orders and the Maintenance Task Checklist subform provide a consistent record for repairs and servicing, and the Maintenance_Request_Notification workflow notifies users when a request is submitted. Parts Inventories, Asset Transfer Management and Service Histories keep the related information in connected forms. Reports, the Tele-ICU Overview page and the dashboards present this information, and the CEO and Manager roles with a data-sharing rule provide controlled access.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Two user roles are configured in the application, as shown in Table 3.1. Technicians are maintained as records in the Technician Management form; whether technicians are also configured as application users is They are not configured as separate application users or assigned a dedicated user role. No additional roles are claimed.

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
CEO	Top of the role hierarchy; the Manager role reports to this role.	Views calibration records shared by the Manager with Read Only access through the "Calibration_Data_Sharing" rule. Other responsibilities:
Manager	Role that reports to the CEO in the role hierarchy.	Creates and maintains records using the configured read and write permissions. Exact forms and actions:

3.2 Functional Requirements

The functional requirements derived from the project objectives are listed in Table 3.2. The FR identifiers are used again in the test cases of Chapter 6. The roles responsible for each requirement are to be confirmed against the application.

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall maintain hospital records and ICU department records.	CEO, Manager	High
FR-02	The system shall maintain technician records.	CEO, Manager	Medium
FR-03	The system shall register a medical asset with an auto-generated Asset ID and the fields of the Medical Asset Registry, linking it to a hospital and an ICU department through lookup fields.	CEO, Manager	High
FR-04	The system shall record the status of an asset as Operational, Under Maintenance, Calibration Due, Out of Service or Retired.	CEO, Manager	High
FR-05	The system shall record calibration schedules, due dates and calibration checklist items.	CEO, Manager	High
FR-06	The system shall identify overdue calibration records and upcoming calibration deadlines using Deluge scripting.	CEO, Manager	High
FR-07	The system shall run a scheduled workflow daily at 9:00 AM that emails calibration deadline alerts.	CEO, Manager	High
FR-08	The system shall allow a maintenance request to be submitted and shall send an email notification on successful submission.	CEO, Manager	High
FR-09	The system shall manage maintenance work orders with a maintenance task checklist.	CEO, Manager	High
FR-10	The system shall maintain a spare-parts inventory.	CEO, Manager	Medium
FR-11	The system shall record medical asset transfers.	CEO, Manager	Medium

FR-12	The system shall maintain service and maintenance history.	CEO, Manager	Medium
FR-13	The system shall provide reports, an overview page and dashboards of asset and calibration information.	CEO, Manager	Medium
FR-14	The system shall share Calibration Management records from the Manager role to the CEO role with Read Only access.	CEO, Manager	High
FR-15	The system shall provide an Approval Request form and an Approval Dashboard.	CEO, Manager	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Target report load time: **5 seconds** for up to **500 records**. *(No performance testing is claimed in this report.)*
NFR-02	Security	Users shall access only the data permitted by their role.
NFR-03	Usability	The application shall be usable through a web browser. Mobile app usage: **available through the Zoho Creator mobile application**.
NFR-04	Availability	The application shall be accessible whenever the Zoho Creator platform is available, subject to internet connectivity and the platform's service availability.
NFR-05	Maintainability	Forms, reports, workflows and scripts shall be maintainable through the platform's configuration tools.
NFR-06	Auditability	Traceability of record changes through the platform's audit facilities

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator — **Professional Edition**
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Not applicable – no external API integration is documented
Other Zoho services	None documented
Notifications	Email
Mobile	Zoho Creator mobile app (**Android and iOS**)
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

The use case diagram of Figure 3.1 covers the CEO and Manager roles. Two key use cases are described in Tables 3.5 and 3.6.

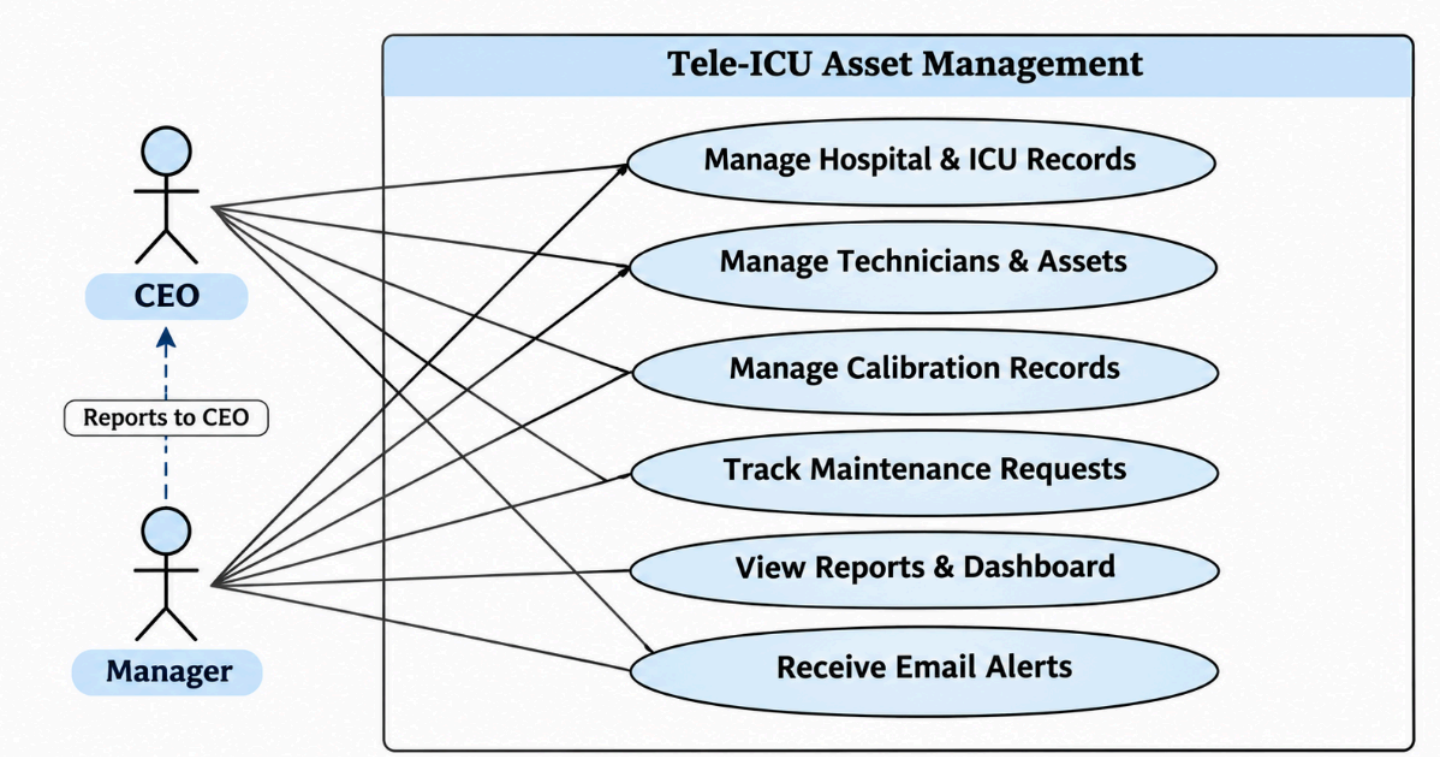


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Monitor Calibration Deadlines

Use Case ID	UC-01
Use Case Name	Monitor Calibration Deadlines
Actor(s)	Manager; CEO (read-only); the system (scheduled workflow)
Pre-condition	Medical asset and calibration records exist with calibration dates recorded.
Main Flow	1. Calibration records and their due dates are held in the application. 2. The Calibration_Deadline_Monitor workflow evaluates the dates using Deluge and identifies overdue and upcoming calibrations, displaying alerts. 3. At 9:00 AM daily, the Daily_Calibration_Reminder workflow checks the records and prepares the reminder content. 4. The reminder email is sent to the recipients.
Alternate Flow	If no calibration is overdue or approaching, no reminder content is generated.
Post-condition	Relevant users are informed of overdue or upcoming calibrations; the CEO can view the shared calibration records.
Related FR	FR-05, FR-06, FR-07, FR-14

Table 3.6: Use Case Description – Submit Maintenance Request

Use Case ID	UC-02
Use Case Name	Submit Maintenance Request
Actor(s)	CEO, Manager
Pre-condition	The medical asset for which maintenance is requested exists in the registry.
Main Flow	1. The user opens the Maintenance Requests form and enters the request details. 2. The user submits the form. 3. On successful submission, the Maintenance_Request_Notification workflow is triggered. 4. An email notification related to the request is sent.
Alternate Flow	If mandatory information is missing, the form is not submitted and no notification is sent. Mandatory fields include Medical Asset, Hospital, Request Date, Issue Description, and Priority.
Post-condition	The request is stored and the notification email is delivered.
Related FR	FR-08

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The architecture of the solution follows the layered structure of a Zoho Creator application. At the top are the users in the CEO and Manager roles, who access the application through a web browser. The application layer consists of the Zoho Creator components: forms and subforms for data capture, reports for viewing and managing records, pages for the overview and dashboards, workflows for automation, and Deluge scripts for logic. Beneath it is the data layer, in which the records submitted through the forms are stored by the platform. The notification layer sends email messages from the workflows. No other Zoho application or external service is integrated in the documented design.

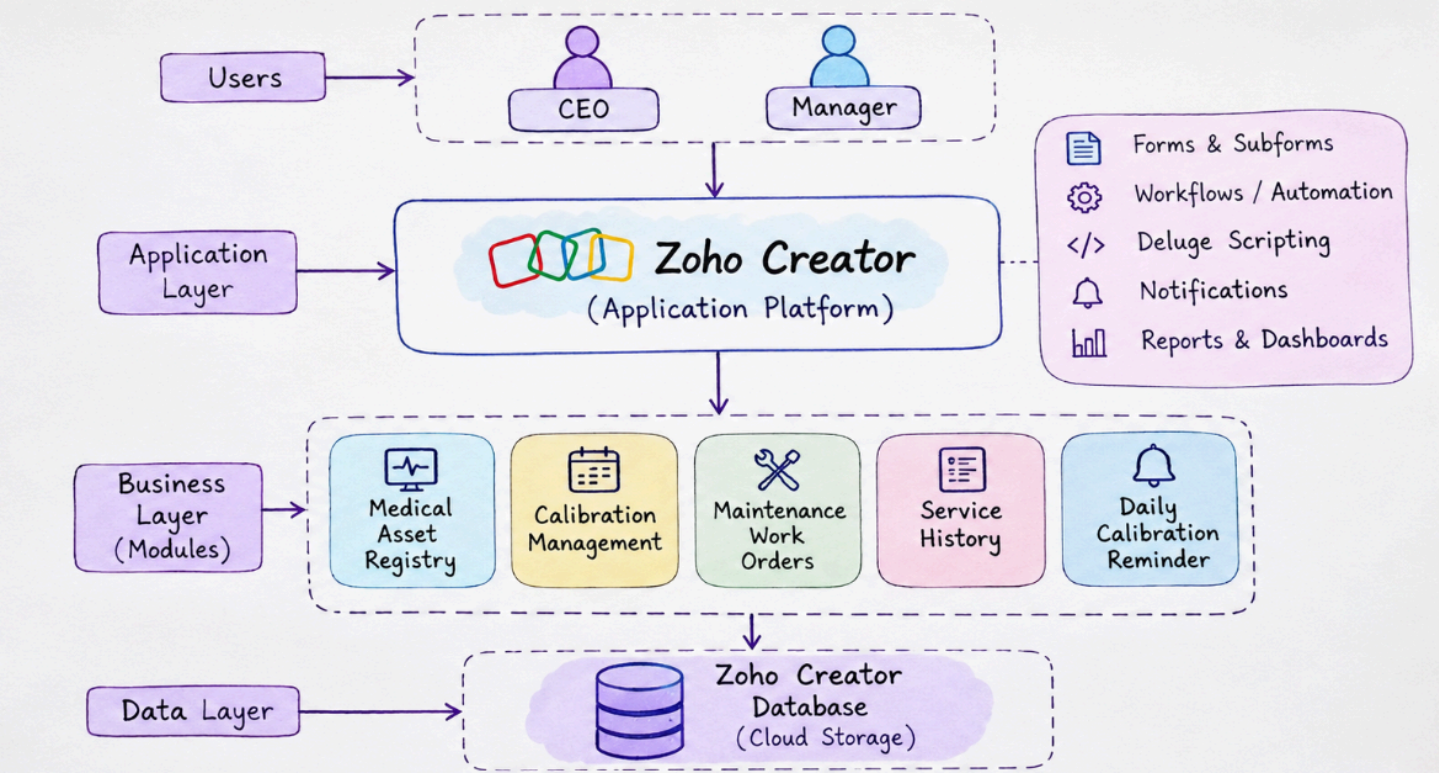


Figure 4.1: System Architecture of the Application

Users and access channel: The CEO and Manager roles sign in to the application and see the forms, reports and pages that their permissions allow. Application layer: Forms capture hospital, department, technician, asset, calibration, maintenance, spare-part, transfer and service-history data; reports and pages present it; workflows react to form events and to a daily schedule. Data layer: Lookup fields relate asset records to hospitals and departments, and subforms hold checklist rows within a parent record. Notification layer: The calibration reminder and the maintenance request notification are delivered by email.

4.2 Data Model

In Zoho Creator, relationships between forms are realised through lookup fields and parent–child relationships through subforms. The relationships confirmed for this application are that the Medical Asset Registry contains

a lookup to Hospital Management (field “Hospital”) and a lookup to ICU Department Management (field “ICU Department Management”), and that the application contains the subforms “Calibration Checklist” and “Maintenance Task Checklist”. Other relationships (for example, between work orders and assets or technicians) and should be added to Figure 4.2 and Table 4.1 from the application.

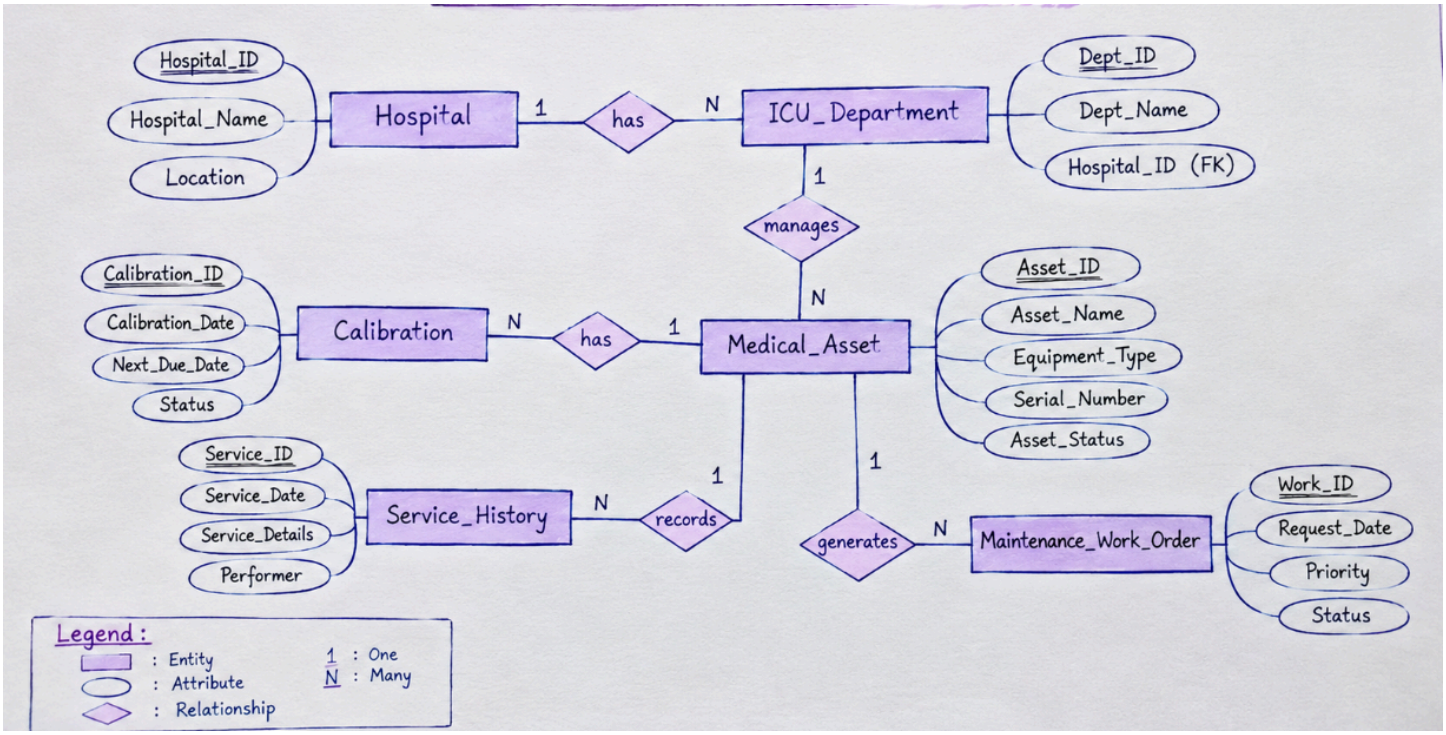


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Hospital Management	Hospital Name, Hospital Code, Address, Contact Number	Medical Asset Registry	One-to-Many	Lookup “Hospital” in Medical Asset Registry
ICU Department Management	Department Name, Department Code, ICU Type, Hospital	Medical Asset Registry	One-to-Many	Lookup “ICU Department Management” in Medical Asset Registry
Medical Asset Registry	Asset ID, Asset Name, Equipment Type, Serial Number, Asset Status, etc. (Table 4.2)	Hospital Management; ICU Department Management	Many-to-One	Lookup fields
Calibration Management	Calibration ID, Medical Asset, Calibration Date, Next Due Date, Status	Calibration Checklist	Parent–Child (to be confirmed)	Lookup field and Calibration Checklist subform
Maintenance Work Orders	Work Order ID, Medical Asset, Assigned Technician, Status, Completion Date	Maintenance Task Checklist	Parent–Child (to be confirmed)	Lookup fields and Maintenance Task Checklist subform
Technician Management; Maintenance Requests; Parts Inventories; Asset Transfer Management; Service Histories; Approval Request	Technician Name, Technician ID, Email, Phone, Specialization.Request ID, Medical Asset, Request Date, Issue Description, Priority, Status	Medical Asset Registry; Hospital Management	Many-to-One	Form records and lookup/reference fields, if configured

4.3 Form Design

Table 4.2 gives the field specification of the Medical Asset Registry form, for which the field list was provided. Field types other than Auto Number and Lookup are inferred from the field names (for example, date fields) and must be confirmed from the form builder together with link names, mandatory status and validations.

Table 4.2: Field Specification – Medical Asset Registry Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Asset ID	Asset_ID	Auto Number	Yes	Automatically generated	System generated
Asset Name	Asset_Name	Single Line	Yes	No duplicate asset names recommended	No duplicate asset names recommended
Equipment Type	Equipment_Type	Drop Down	Yes	Ventilator, ECG Machine, etc.	Sample value: Ventilator
Manufacturer	Manufacturer	Single Line	Yes	-	Sample value: Philips
Model Number	Model_Number	Single Line	Yes	-	Sample value: V60
Serial Number	Serial Number	Single Line	Yes	Unique serial number	Sample value: PH-VENT-002
Hospital	Hospital	Lookup	Yes	Lookup to Hospital Management	Lookup to Hospital Management
ICU Department Management	ICU_Department_Management	Lookup	Yes	Lookup to ICU Department Management	Lookup to ICU Department Management
Installation Date	Installation_Date	Date (confirm)	Yes	Date must not be in the future	-
Last Calibration Date	Last_Calibration_Date	Date (confirm)	No	Must be on or before the Next Calibration Due date	-
Next Calibration Due	Next_Calibration_Due	Date (confirm)	Yes	-	-
Asset Status	Asset_Status	Drop-down / choice (confirm)	No	Operational, Under Maintenance,	Operational, Under Maintenance, Calibration Due, Out of Service, Retired

				Calibration Due, Out of Service, Retired	
Purchase Cost	Purchase_Cost		No		
Warranty Expiry Date	Warranty_Expiry_Date	Date (confirm)	No	Must be zero or a positive value	
Equipment Image	Equipment_Image	Image (confirm)	-		
Equipment Documents	Equipment_Documents	File upload (confirm)	-		

4.4 Process and Workflow Design

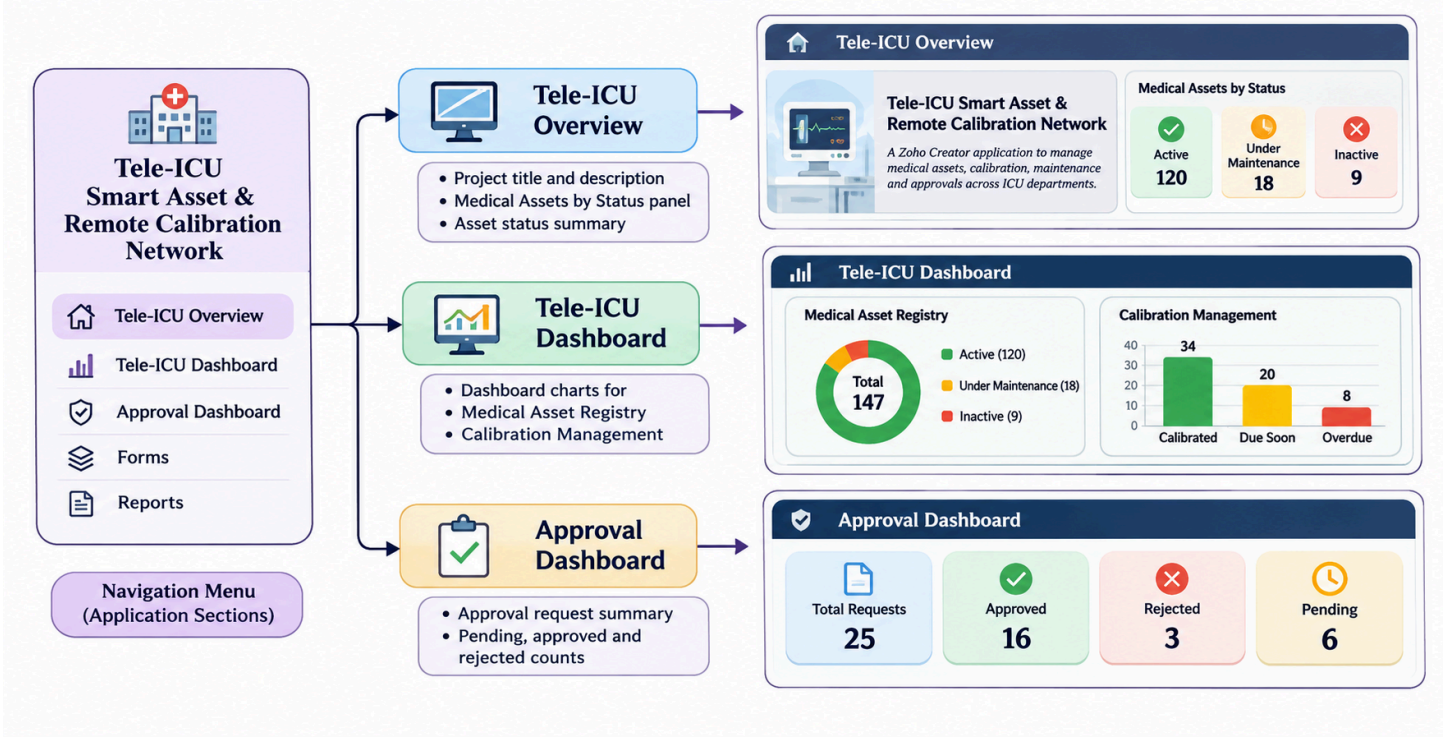
Two automated processes are designed. The calibration monitoring process begins with the calibration dates held in the records. The Calibration_Deadline_Monitor workflow uses Deluge to compare the due dates with the current date, identifies overdue and upcoming records and displays alerts. The Daily_Calibration_Reminder workflow, scheduled at 9:00 AM daily, checks the records for overdue or upcoming deadlines, prepares the email content with Deluge and sends the alert email. The maintenance notification process begins when a user submits a maintenance request; on successful submission, the Maintenance_Request_Notification workflow sends an email notification related to the request.

Figure 4.3: Workflow / Process Flow for Calibration Monitoring and Maintenance Request Notification

4.5 User Interface and Navigation Design

The application presents three pages: the Tele-ICU Overview, which shows the project title and description, a Medical Assets by Status panel and an asset-status summary; the Tele-ICU Dashboard, which shows dashboard charts for the Medical Asset Registry and Calibration Management; and an Approval Dashboard. The forms and their reports are organised as application sections.

Figure 4.4: Navigation Structure and Wireframe of the Tele-ICU Overview Page



4.6 Access Control Design

Access control is based on two roles, CEO and Manager, arranged in a hierarchy in which the Manager reports to the CEO. Read and write permissions are configured. A data-sharing rule named “Calibration_Data_Sharing” shares records of Calibration Management from the Manager role to the CEO role with Read Only access. Table 4.3 records the permission matrix; the per-component permissions (A = Add, E = Edit, V = View, D = Delete, — = No access) must be completed from the application, and only the sharing rule is stated here as configured.

Table 4.3: Role–Permission Matrix

Component	CEO	Manager
Form: Calibration Management	V (Read Only, via Calibration_Data_Sharing for Manager-owned records)	
Form: Medical Asset Registry		
Form: Maintenance Requests		
Form: Maintenance Work Orders		
Other forms (Hospital, ICU Department, Technician, Parts Inventories, Asset Transfer, Service Histories, Approval Request)		
Reports		
Pages: Tele-ICU Overview, Tele-ICU Dashboard, Approval Dashboard		

CHAPTER 5

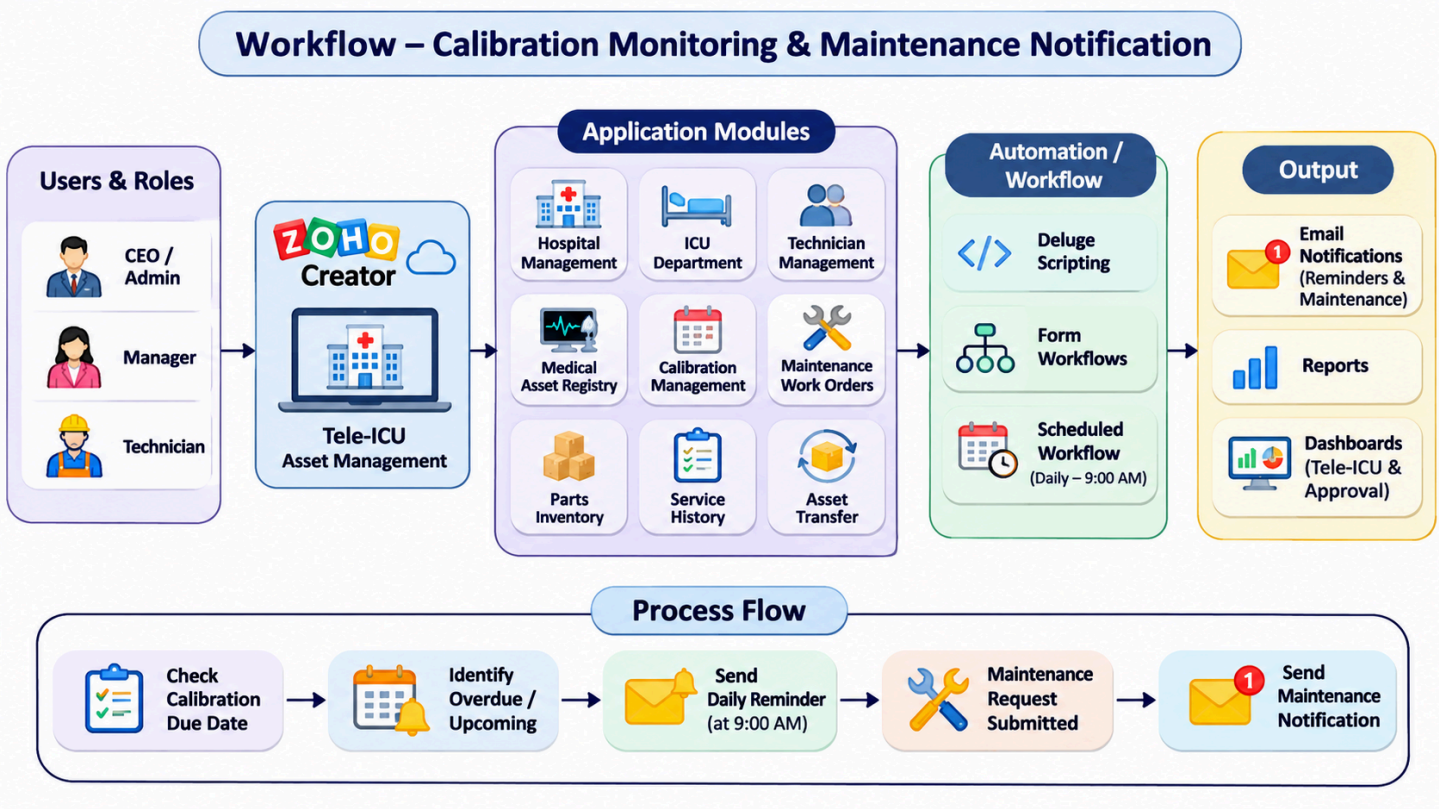
IMPLEMENTATION

This chapter describes the implementation of the Tele-ICU Asset Management application on Zoho Creator. For each component, the purpose and configuration are described and a screenshot placeholder is provided. Only features documented for the application are described; items whose details were not available are marked for completion.

5.1 Application Setup

The application was created on the Zoho Creator platform under the name “Tele-ICU Asset Management”. The method of creation (from scratch, template, import or AI-assisted builder), the workspace, the Zoho Creator edition and the application link name are . The application is organised around the forms of the equipment-management process (Section 5.2), their reports (Section 5.3) and the pages described in Section 5.4. The names of the application sections and menu items, and any theme or branding applied.

Figure 5.1: Application Home and Navigation Sections



5.2 Forms and Data Capture

Forms are the means by which data is captured and stored in Zoho Creator. The following forms and modules have been created or documented for the application. The exact final number of forms is to be confirmed from the application.

- Hospital Management – records hospitals. A sample record named “Chennai Central Hospital” with the code HSP001 was used for testing.
- ICU Department Management – records ICU departments of the hospitals.
- Technician Management – records technician information.
- Medical Asset Registry – the central registry of medical equipment (described below).
- Calibration Management – records calibration schedules and due dates and holds the Calibration Checklist subform (host form to be confirmed).
- Maintenance Requests – used to submit and track maintenance requests; successful submission triggers an email notification.
- Maintenance Work Orders – manages maintenance work orders and holds the Maintenance Task Checklist subform (host form to be confirmed).
- Parts Inventories (Spare Parts Inventory) – maintains spare-parts inventory.
- Asset Transfer Management – records the transfer of medical assets.
- Service Histories – maintains the service and maintenance history of assets.
- Approval Request – supports approval-related activity; its role is stated here only as far as the application supports it.

5.2.1 Medical Asset Registry

The Medical Asset Registry is the principal form of the application. It contains the fields Asset ID (Auto Number), Asset Name, Equipment Type, Manufacturer, Model Number, Serial Number, Hospital (Lookup), ICU Department Management (Lookup), Installation Date, Last Calibration Date, Next Calibration Due, Asset Status, Purchase Cost, Warranty Expiry Date, Equipment Image and Equipment Documents. The Asset Status field offers the choices Operational, Under Maintenance, Calibration Due, Out of Service and Retired. The Hospital and ICU Department Management lookups relate each asset to the hospital and department in which it is located; during testing the lookups worked when the corresponding hospital and department records existed.

A sample test record used during development had the equipment type Ventilator, manufacturer Philips, model number V60, serial number PH-VENT-002, hospital Chennai Central Hospital, asset status Operational, installation date 01-Oct-2026, last calibration date 01-Oct-2026 and next calibration due date 01-Nov-2026. Two sample assets, “ICU Ventilator 001” and “ICU Ventilator 002”, were created. These are sample test records and not real hospital data.

5.2.2 Subforms

Two subforms are used. The Calibration Checklist subform records calibration checklist items and related details, and the Maintenance Task Checklist subform records maintenance tasks and their completion details. A subform allows several rows of related data to be entered within one parent record. The fields of each subform, and the forms in which they are embedded.

5.2.3 Field Validations and Form Properties

Field-level validations, layout rules, success messages and other form properties are . No validation or layout rule is claimed in this report unless it is added here by the student.

Figure 5.2: Medical Asset Registry Form in the Form Builder

Figure 5.3: Medical Asset Registry Form in Live Mode

5.3 Reports

Reports have been created and are available in the application for viewing and managing the records of the forms. The reports named for this report are the Medical Asset Registry Report, All Hospital Management and All Calibration Management, together with other reports corresponding to the created forms. The exact number of reports and their complete names are [TO BE FILLED BY STUDENT]; Table 5.1 lists the known reports. Report types, filters, grouping, sorting, conditional formatting and custom actions are to be completed from the application.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Medical Asset Registry Report	[TO BE FILLED BY STUDENT]	Medical Asset Registry	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]
All Hospital Management	[TO BE FILLED BY STUDENT]	Hospital Management	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]
All Calibration Management	[TO BE FILLED BY STUDENT]	Calibration Management	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]
Reports for the remaining forms	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]

[SCREENSHOT TO BE ADDED: Medical Asset Registry Report]

Figure 5.4: Medical Asset Registry Report

5.4 Pages and Dashboards

Pages built with the UI builder present summaries and visualisations of the application data. Three pages are documented.

- Tele-ICU Overview – shows the project title and description, a Medical Assets by Status panel and an asset-status summary. During testing, asset status information was displayed in this overview.
- Tele-ICU Dashboard – shows dashboard charts for the Medical Asset Registry and for Calibration Management. During testing, the charts displayed application data. The chart types and the fields plotted are [TO BE FILLED BY STUDENT – SCREENSHOT / DETAILS TO BE ADDED].
- Approval Dashboard – an approval-related dashboard available in the application; its components are [TO BE FILLED BY STUDENT – SCREENSHOT / DETAILS TO BE ADDED].

[SCREENSHOT TO BE ADDED: Tele-ICU Overview page]

[SCREENSHOT TO BE ADDED: Tele-ICU Dashboard page]

Figure 5.5: Tele-ICU Overview and Tele-ICU Dashboard Pages

5.5 Deluge Scripting

Deluge is the scripting language of Zoho Creator [9]. In this application it is used to check calibration due dates, identify overdue calibration records, identify upcoming calibration deadlines, prepare email content dynamically, send calibration reminder emails, and support maintenance request email notifications. Table 5.2 lists the scripts. The exact source code has not been supplied for this report; therefore the code boxes below contain placeholders, and no code has been written on the student's behalf.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Workflow "Calibration_Deadline_Monitor" – trigger [TO BE FILLED BY STUDENT]	Check calibration records and due dates; identify overdue and upcoming calibrations; display alerts	Date evaluation, alerts. Statements used: [TO BE FILLED BY STUDENT]
DS-02	Scheduled workflow "Daily_Calibration_Reminder" – daily at 9:00 AM	Check for overdue or upcoming deadlines; prepare detailed email content; send calibration alert emails	Date evaluation, dynamic message construction, email sending. Statements used: [TO BE FILLED BY STUDENT]
DS-03	Workflow "Maintenance_Request_Notification" – on successful submission of a maintenance request	Support the email notification related to a maintenance request	Email sending. Statements used: [TO BE FILLED BY STUDENT]

Code 5.1: Calibration deadline monitoring script (DS-01)

```
// Script ID : DS-01

// Location  : Workflow Calibration_Deadline_Monitor

// Purpose   : Identify overdue and upcoming calibration records; display alerts

[PASTE EXACT DELUGE CODE HERE – TO BE FILLED BY STUDENT]
```

Code 5.2: Daily calibration reminder script (DS-02)

```
// Script ID : DS-02

// Location  : Scheduled workflow Daily_Calibration_Reminder (daily, 9:00 AM)

// Purpose   : Prepare calibration reminder content and send the email

[PASTE EXACT DELUGE CODE HERE - TO BE FILLED BY STUDENT]
```

Code 5.3: Maintenance request notification script (DS-03)

```
// Script ID : DS-03

// Location  : Workflow Maintenance_Request_Notification (on successful submission)

// Purpose   : Send the email notification related to a maintenance request

[PASTE EXACT DELUGE CODE HERE - TO BE FILLED BY STUDENT]
```

Logic and results: The calibration deadline script compares the recorded calibration due dates with the current date to classify records as overdue or upcoming. The reminder script applies the same check, builds the content of the reminder email from the records found and sends it. When the script was executed manually it ran successfully and the reminder email was received. The notification script sends an email when a maintenance request is submitted; with a test request the email was received. [TO BE FILLED BY STUDENT – describe the actual logic of each script once the code is pasted, for example the number of days used to define “upcoming”.]

5.6 Workflows and Automation

Three workflows are documented, as listed in Table 5.3. The Calibration_Deadline_Monitor workflow checks calibration records and due dates and uses Deluge to evaluate dates and display alerts. The Daily_Calibration_Reminder workflow is a scheduled workflow configured to run daily at 9:00 AM; it checks calibration records for overdue or upcoming deadlines, prepares the reminder content with Deluge and sends the alert email. The script was executed manually and the email was received, and the scheduled workflow is enabled; an independent scheduled execution has not been verified in this report. The Maintenance_Request_Notification workflow is triggered when a maintenance request is successfully submitted and sends an email notification; a test request was submitted and the notification was received.

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Calibration_Deadline_Monitor	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	Overdue or upcoming calibration due dates	Evaluate dates; display alerts	DS-01
Daily_Calibration_Reminder	Schedule	Daily at 9:00 AM	Overdue or upcoming calibration deadlines	Prepare and send calibration alert email	DS-02
Maintenance_Request_Notification	Form workflow (confirm)	Successful submission of a maintenance request	[TO BE FILLED BY STUDENT]	Send email notification	DS-03

Approval processes, Blueprints and batch or report workflows are not documented for this application. An Approval Request form and an Approval Dashboard exist; whether an approval process is configured is [TO BE FILLED BY STUDENT].

[SCREENSHOT TO BE ADDED: Workflow configuration – Calibration_Deadline_Monitor, Daily_Calibration_Reminder and Maintenance_Request_Notification]

Figure 5.6: Workflow Configuration

5.7 Integrations

No integration with another Zoho service, external API, webhook or Creator REST API is documented for this application, and no connection or invokeurl call is used. The only outward communication is email notification from the workflows, as summarised in Table 5.4. Consequently no integration script is included and Postman testing is not applicable. If an integration exists in the actual application, it is [TO BE FILLED BY STUDENT].

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Email notifications	Email	Sent by workflows and Deluge	Not applicable	Calibration reminders; maintenance request notifications
Zoho / external services	None documented	Not applicable	Not applicable	Not applicable

[SCREENSHOT TO BE ADDED: Email notification configuration and the calibration reminder / maintenance request emails received during testing]

Figure 5.7: Email Notification Configuration and Received Emails

5.8 User Management and Role-Based Access

Role-based access is implemented through two roles, CEO and Manager. The Manager role reports to the CEO in the role hierarchy. Read permission and write permission are configured. A data-sharing rule named “Calibration_Data_Sharing” was configured for the Calibration Management form; it shares records from the Manager role to the CEO role with Read Only access, so that the CEO can view the calibration records of the Manager without modifying them. No additional roles, permission profiles or record-level criteria are claimed. The users assigned to each role and the permission profile settings are [TO BE FILLED BY STUDENT].

[SCREENSHOT TO BE ADDED: Roles hierarchy, permissions and the Calibration_Data_Sharing rule]

Figure 5.8: Roles and Permission Profiles Configuration

5.9 AI and Analytics Features

No Zia or other AI feature is documented for this application. [TO BE FILLED BY STUDENT – if any AI-assisted feature or AI-generated script was used, state it here and how it was verified, as required by the template.] Analytics are provided through the Tele-ICU Overview, which includes a Medical Assets by Status panel and an asset-status summary, and through the Tele-ICU Dashboard, which has charts for the Medical Asset Registry and Calibration Management. These present the current data of the application; during testing the charts displayed application data. Integration with Zoho Analytics is not documented.

[SCREENSHOT TO BE ADDED: Tele-ICU Dashboard charts / Medical Assets by Status panel]

Figure 5.9: Analytics Features in Use

5.10 Pro-Code Extension

No pro-code extension (Zoho Catalyst, custom JavaScript widgets or Creator REST API clients) is documented for this application. The requirements of equipment, calibration and maintenance record management were met by forms, reports, pages, workflows and Deluge scripting, so a pro-code component was not necessary. Such an extension could be considered in the future if the application is to exchange data with external hospital systems or to perform heavy processing that exceeds the limits of Deluge. [TO BE FILLED BY STUDENT – confirm that no pro-code extension was used.]

5.11 Mobile and Portal Access

Access through the Zoho Creator mobile app and a customer portal is optional in the template. No customer portal is part of the project. Whether the application was viewed on the mobile app is [TO BE FILLED BY STUDENT].

[SCREENSHOT TO BE ADDED: Application on mobile (if used)]

Figure 5.10: Application on Mobile

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out functionally during development using sample test records. At the level of individual components, medical asset records were created and viewed in reports, and the lookup fields were exercised. At the level of scripts, the calibration deadline Deluge script was executed manually. At the level of workflows, the calibration reminder was produced by manual execution and the maintenance request notification was tested by submitting a test request; in both cases the email was received. The overview page and dashboard charts were checked for the display of application data. Role-based access testing (logging in as each role), negative testing, independent verification of the scheduled run, and user acceptance testing are not recorded in this report and are marked for the student to complete if they were performed. Performance testing and security testing are not claimed.

6.2 Test Cases

Table 6.1 lists the test cases. Cases that were verified during development are marked Pass. For the remaining cases the expected result is stated, but the actual result and status are left to be filled in after execution; no result has been assumed.

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-03	Create a medical asset record	Enter sample asset data in the Medical Asset Registry form and submit	Record created and visible in the Medical Asset Registry Report	Records created and viewed in reports	Pass
TC-02	FR-03	Lookup fields with existing records	Create hospital and ICU department records, then select them in the asset form	Hospital and department can be selected through the lookups	Lookups worked when records existed	Pass
TC-03	FR-06	Execute calibration deadline script	Run the Deluge script manually	Script executes without error	Executed successfully	Pass
TC-04	FR-07	Calibration reminder email (manual run)	Execute the reminder script manually	Reminder email received	Email received	Pass
TC-05	FR-07	Scheduled daily run at 9:00 AM	Allow the enabled schedule to run unaided	Reminder email received at 9:00 AM	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]
TC-06	FR-08	Submit a maintenance request	Submit a test request in the Maintenance Requests form	Request saved	Submission tested	Pass
TC-07	FR-08	Maintenance request notification	Submit a test request	Notification email received	Email received	Pass
TC-08	FR-04	Asset status in overview	Open the Tele-ICU Overview page	Asset status information displayed	Displayed	Pass
TC-09	FR-13	Reports available	Open the reports of the application	Reports list the records	Reports available	Pass
TC-10	FR-13	Dashboard charts	Open the Tele-ICU	Charts display	Displayed data	Pass

			Dashboard	application data		
TC-11	FR-14	CEO views Manager calibration records	Log in as CEO and open Calibration Management	Records visible, Read Only	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]
TC-12	NFR-02	Unauthorised access (negative)	Log in as each role and attempt actions outside its permission	Access denied	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]
TC-13	FR-08	Maintenance request with missing mandatory data (negative)	Submit without a mandatory field	Validation error; no email	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]
TC-14	FR-05	Calibration checklist rows	Add rows in the Calibration Checklist subform	Rows saved with the parent record	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]
TC-15	FR-09	Work order with task checklist	Create a work order and add task checklist rows	Work order saved with tasks	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]
TC-16	FR-10, 11, 12	Spare part, asset transfer and service history	Enter one record in each form	Records saved	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]
TC-17	FR-15	Approval Request	Submit an approval request	As configured: [TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]

6.3 Debugging Deluge Scripts

Deluge scripts in Zoho Creator can be examined by executing them from the script editor, by inspecting the values of variables with info statements, and by reading the error messages shown when a form is submitted or a workflow runs. In this project the calibration reminder script was executed manually to confirm its behaviour before relying on the schedule. A description of one real bug encountered during development, its cause and the fix is [TO BE FILLED BY STUDENT]; none is reported here because none was provided.

[SCREENSHOT TO BE ADDED: Deluge execution output / log]

Figure 6.1: Deluge Debug Output / Log

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]
BUG-02	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]

6.5 User Acceptance Testing

No user acceptance testing is recorded for this project. If feedback was collected from at least two end users using the application in an assigned role, it should be entered in Table 6.3.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]
[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The core application has been developed and tested using sample records. The use of the Development, Stage and Production environments, the publishing of the application, the final application URL, version notes and the rollback plan are [TO BE FILLED BY STUDENT]. This report does not claim deployment to production or production use.

[SCREENSHOT TO BE ADDED: Environments / Publish to Production (if used)]

Figure 7.1: Environments / Publish to Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	Partly (Table 6.1) – [TO BE FILLED BY STUDENT]
2	Sample/test data removed or isolated	[TO BE FILLED BY STUDENT]
3	Roles, permissions and users configured in Production	[TO BE FILLED BY STUDENT]
4	Connections authorised in Production	Not applicable (no connections documented)
5	Schedules and workflows enabled	Scheduled workflow enabled; others [TO BE FILLED BY STUDENT]
6	Application published and URL verified	[TO BE FILLED BY STUDENT]
7	Version / release notes recorded	[TO BE FILLED BY STUDENT]

7.2 Performance Optimisation

No performance testing or optimisation measurement was carried out or is claimed for this application. Table 7.2 is retained from the template for the student to complete if any optimisation, such as using filtered reports or efficient criteria in scripts, was applied and measured.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]

7.3 Security Features

The security measures documented for the application are the role hierarchy with read and write permissions, and the Read Only data-sharing rule for Calibration Management. Table 7.3 lists them. Other controls such as encrypted fields, field-level hiding, IP or domain restrictions, two-factor authentication and OAuth connections are not documented and are not claimed. No security testing is claimed.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role hierarchy with read and write permissions	CEO and Manager roles	Unauthorised access to or modification of data
Read Only data-sharing rule “Calibration_Data_Sharing”	Calibration Management (Manager to CEO)	Unintended modification of calibration records by the viewing role
[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]	[TO BE FILLED BY STUDENT]

7.4 Governance and Administration

Governance of the application includes keeping naming conventions consistent (for example, the workflow names Calibration_Deadline_Monitor, Daily_Calibration_Reminder and Maintenance_Request_Notification), controlling who can change data through roles and permissions, and keeping the scheduled workflow and email recipients up to date. The use of the audit trail, change history, backup and export, user licence management and maintenance responsibilities are [TO BE FILLED BY STUDENT].

[SCREENSHOT TO BE ADDED: Audit Trail / Administration Settings (if used)]

Figure 7.2: Audit Trail / Administration Settings

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Application Output

This section follows a realistic scenario through the application using sample test records. The records are for testing only and do not represent real hospital data.

Step 1 – Registering the hospital and the asset. A sample hospital, “Chennai Central Hospital” (code HSP001), is recorded in Hospital Management. A medical asset is then registered in the Medical Asset Registry with the equipment type Ventilator, manufacturer Philips, model number V60, serial number PH-VENT-002, the hospital selected through the lookup, asset status Operational, installation date 01-Oct-2026, last calibration date 01-Oct-2026 and next calibration due date 01-Nov-2026. The record appears in the Medical Asset Registry Report, and the asset-status information is reflected in the Tele-ICU Overview.

[SCREENSHOT TO BE ADDED: Step 1 – Hospital and medical asset record created and visible in the report]

Figure 8.1: Scenario Step 1 – Registering a Hospital and a Medical Asset

Step 2 – Calibration deadline monitoring. The calibration deadline script evaluates the calibration due dates and identifies overdue or upcoming records. The Daily_Calibration_Reminder script prepares the reminder content and sends the email; when executed manually, the script ran successfully and the calibration reminder email was received. The scheduled workflow is enabled to run daily at 9:00 AM.

[SCREENSHOT TO BE ADDED: Step 2 – Calibration reminder email received]

Figure 8.2: Scenario Step 2 – Calibration Deadline Reminder

Step 3 – Submitting a maintenance request. A test maintenance request is submitted in the Maintenance Requests form. On successful submission, the Maintenance_Request_Notification workflow sends an email notification, which was received during testing. The Tele-ICU Dashboard presents the charts for the Medical Asset Registry and Calibration Management using the application data.

[SCREENSHOT TO BE ADDED: Step 3 – Maintenance request submitted and notification email received]

Figure 8.3: Scenario Step 3 – Maintenance Request and Notification

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
1. Centralised medical asset registry with hospital, department and technician records	Sections 5.2.1, 5.2; Hospital Management, ICU Department Management, Technician Management and Medical Asset Registry forms	Asset records created and viewed in reports; lookups worked with existing records (TC-01, TC-02)
2. Calibration schedules, due dates and checklists	Calibration Management, Calibration Checklist subform, Next Calibration Due field	Calibration fields available; checklist subform testing
3. Maintenance requests, work orders and task checklists	Maintenance Requests, Maintenance Work Orders, Maintenance Task Checklist subform	Request submission tested (TC-06); work-order testing
4. Spare parts, asset transfers, service history	Parts Inventories, Asset Transfer Management, Service Histories	Testing
5. Automated deadline monitoring and email notifications	Sections 5.5, 5.6; DS-01 to DS-03	Script executed; reminder and notification emails received (TC-03, TC-04, TC-07)
6. Reports, dashboards, roles and permissions	Sections 5.3, 5.4, 5.8	Reports and dashboard charts available (TC-09, TC-10); role testing

8.3 Comparison with the Existing System

Table 8.2 compares the existing manual or disconnected process with the proposed application. Quantitative improvements have not been measured in this project and are left for the student to complete if measurements are made.

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Time to process one request	Depends on manual communication	Request recorded and notified by email on submission	
Manual steps	Dates tracked and reminders issued manually	Deadline checks and reminder emails automated by workflows and Deluge	
Report generation	Compiled from separate records	Reports and dashboards built from the stored records	
Data accessibility	Records scattered across registers or files	Centralised records with role-based access	

8.4 Limitations of the Low-Code Approach

The limitations of the project itself are as follows. The application has been verified only with sample test records and functional tests; performance, security and user acceptance testing have not been carried out or recorded. The calibration reminder has been executed manually and the schedule has not been independently observed. The application does not communicate with the medical devices, and it uses email as its only notification channel.

As regards the low-code approach in general, practitioners hold conflicting views on its advantages and disadvantages [4]. Typical constraints include dependence on the features and limits of the chosen platform, limited control over the user interface beyond what the platform offers, the need for scripting when logic goes beyond configuration, and the difficulty of moving an application to another platform. The specific constraints encountered on Zoho Creator during this project, and the workarounds adopted.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The parts of the application can be placed on the spectrum as follows. The forms, field types, lookups, subforms, reports, pages, dashboard charts, and the role hierarchy and data-sharing rule were produced through configuration, which corresponds to the no-code end. The calibration date evaluation and the dynamic construction and sending of the reminder email needed Deluge scripting, which corresponds to the low-code level. No pro-code component was used. Pro-code development would be considered if the application had to exchange data with medical devices or external hospital systems at a scale or in a form that the platform cannot support through configuration and scripting.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Tele-ICU Smart Asset & Remote Calibration Network was developed as the low-code application “Tele-ICU Asset Management” on Zoho Creator to centralise the management of medical equipment across hospitals and ICU departments. The application provides forms for hospitals, ICU departments, technicians, medical assets, calibration management, maintenance requests, work orders, spare parts, asset transfers, service histories and approval requests, with subforms for calibration and maintenance checklists. Deluge scripting and workflows monitor calibration deadlines, send a daily calibration reminder and notify users of maintenance requests by email. Reports, an overview page and dashboards present the data, and a CEO–Manager role hierarchy with a read-only data-sharing rule controls access to calibration records.

Functional testing with sample records confirmed that asset records can be created and reported, lookups work, the calibration script executes, and the reminder and notification emails are delivered. Deployment, performance, security and user acceptance evaluations remain to be completed and are marked in this report. The project shows that a coherent equipment-management solution can be built on a low-code platform with a small amount of scripting.

9.2 Future Enhancements

- Complete and record role-based, negative, performance and user acceptance testing, and observe the scheduled reminder over several days.
- Extend the notifications and dashboards, for example with additional reports for spare-parts stock and service history.
- Consider integration with other systems through Zoho Creator’s integration capabilities, or a pro-code extension, if exchange of data with external hospital systems becomes necessary.
-

9.3 Learning Outcomes and Career Relevance

Developing this project involved data modelling with forms, lookups and subforms, building reports and dashboards, writing Deluge scripts and workflows, and configuring roles and permissions on a real platform. These skills relate to roles such as citizen developer, low-code developer, solutions consultant and automation engineer.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Each feature is marked according to what is documented for the application. Entries marked must be verified against the live application, since examiners check this table during the demonstration.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y (sections:	Tele-ICU Asset Management	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y / N	Medical Asset Registry and others	5.2	
II	Lookup field (relationship between forms)	Y	Medical Asset Registry – Hospital, ICU Department Management	4.2, 5.2	Table 4.1
II	Subform (parent–child data)	Y	Calibration Checklist; Maintenance Task Checklist	4.2, 5.2	
II	Field validations / layout rules	Y / N		5.2	
II	List report with filter, grouping, custom action	Y		5.3	Table 5.1
II	Kanban / Calendar / Timeline / Map report	Y / N –		5.3	
II	Pivot table / chart report	Y / N –		5.3	
II	Page / dashboard built with UI Builder	Y	Tele-ICU Overview; Tele-ICU Dashboard; Approval Dashboard	5.4	Fig. 5.5
II	Deluge – form events (validate / success / field actions)	Y / N –		5.5	
II	Deluge – functions, fetch, loops, collections	Y / N		5.5	

II	Email / in-app notifications	Y (email)	Daily_Calibration_Reminder; Maintenance_Request_Notification	5.5, 5.6	Fig. 5.7
II	Form workflow	Y (confirm)	Maintenance_Request_Notification	5.6	Table 5.3
II	Scheduled workflow	Y	Daily_Calibration_Reminder (daily 9:00 AM)	5.6	Table 5.3
II	Approval process	Y / N –	Approval Request form; Approval Dashboard	5.6	
II	Blueprint	N (not documented)	–	5.6	–
II	Batch / report workflow	N (not documented)	–	5.6	–
II	Testing & debugging evidence	Y (partial)	Table 6.1	6	Table 6.1
III	Integration with a Zoho service	N (not documented)	–	5.7	–
III	Connection / REST API call (invokeurl)	N (not documented)	–	5.7	–
III	Webhook or Creator REST API	N (not documented)	–	5.7	–
III	Users, roles and permission profiles	Y (roles: CEO, Manager)	Role hierarchy; permissions	5.8	Fig. 5.8
III	Record-level access criteria / customer portal	Partly – data-sharing rule; portal N	Calibration_Data_Sharing	5.8	Fig. 5.8
III	Zia / AI feature	N (not documented)	–	5.9	–

III	Analytics (charts / Zoho Analytics)	Y (dashboard charts)	Tele-ICU Dashboard	5.9	Fig. 5.9
IV	Performance optimisation applied & measured	N (not documented)	–	7.2	–
IV	Deployment across environments	Y / N –	-	7.1	
IV	Security controls (encryption, OAuth, permissions)	Partly – permissions	Roles; sharing rule	7.3	Table 7.3
IV	Governance (audit trail, administration)	Y / N –	Mangement	7.4	
V	Pro-code extension (Catalyst / widgets / custom functions)	N (not documented)	–	5.10	–
V	Low-code limitations & spectrum analysis	Y	Report	8.4, 8.5	–
—	Mobile app / portal access	Y / N –	Report	5.11	

Total features implemented: 30/ 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	4	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	3	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	4	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	4	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

The exact Deluge code was not supplied for this report. Paste each script in full in the boxes below (Courier New, 10 pt), as text and not as screenshots.

C.1 Calibration_Deadline_Monitor

```
if(input.Next_Due_Date != null)

{

    if(input.Next_Due_Date < zoho.currentdate)

    {

        info "ALERT: Calibration is overdue!";

    }

    else if(input.Next_Due_Date <= zoho.currentdate.addDay(7))

    {

        info "WARNING: Calibration is due within 7 days.";

    }

    else

    {

        info "Calibration is not due yet.";

    }

}
```

MANUAL

D.1 Accessing the Application

- Open a supported web browser (Google Chrome, Microsoft Edge or Mozilla Firefox).
- Open the application :
- Sign in with the credentials assigned to your role (CEO or Manager).
- Select the application “Tele-ICU Asset Management”.

D.2 Instructions for the Manager

- Add hospital and ICU department records before creating assets, so that the lookups in the Medical Asset Registry have values to select.

- Open the Medical Asset Registry form, enter the asset details, select the hospital and ICU department, set the asset status and calibration dates, and submit.
- Maintain calibration records in Calibration Management, including checklist rows.
- Submit maintenance requests and manage work orders, spare parts, asset transfers and service histories through their forms.
- Review reports and the Tele-ICU Dashboard for asset and calibration information. Calibration reminder emails are sent daily at 9:00 AM.

D.3 Instructions for the CEO

- Sign in and open the Tele-ICU Overview and Tele-ICU Dashboard to view asset status and calibration charts.
- Open Calibration Management to view the records shared by the Manager; access to these records is Read Only.

-

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	Shremathi K	Register No.	2116240701504
Project Title	Tele-ICU Smart Asset & Remote Calibration Network	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	Shremathi K	Register No.	2116240701504
Department / Section	Computer Science and Engineering	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER – Name: _____

EXTERNAL EXAMINER – Name: _____